

Harsh Gupta

Embedded Systems & IoT Firmware Engineer • 5 Years Experience • MERIQ Index: 71% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: ARM Cortex, C/C++, FreeRTOS, PCB Debugging, Python, I2C/SPI/UART, Embedded Linux, MQTT

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core ARM Cortex service, reducing latency by 29%.
- Led a squad of 7 engineers delivering scalable microservices processing 50M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 56%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2021 - 2023)

- Designed and maintained C/C++ data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using FreeRTOS saving \$10k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [ARM Cortex, C/C++, FreeRTOS, PCB Debugging]

Engineered end-to-end production solution leveraging ARM Cortex and C/C++ with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [FreeRTOS, PCB Debugging, Python, I2C/SPI/UART]

Architected high-resilience system integrating FreeRTOS with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

Stanford University — 2017 - 2021 (Score: 8.26 / 10.0)